

**PROGRAMME: BCA (Bachelor of Computer Applications)
SEMESTER – VI**

P2: Teaching-Learning and Evaluation Plan

PROGRAMME: BCA (6th Semester ISMA & MACT)

Teaching - Learning & Evaluation Plan

Course Information: Core Course

Course Code: **23BCA6CTE02** Course Title: **AMAZON WEB SERVICES** Credits Units: **03**

Contact Hours: **45** L-T-P: **3-0-0**

IA: ESE Weightage – **50:50**

ESE Question Paper Marks: **50**

Pre-requisite (if any):

Pass Marks (IA & ESE) –**NA**

:18 Special Examination

Fees: **NA**

Aggregate Pass Marks: **40**

Students should have completed course on Cloud Computing & Management of Data Centres

Course Facilitator (s): Dr Manivasagam G

Programme Outcomes (POs)	
At the end of the programme, students will be able to	
PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand, and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO	Life-long Learning: Recognize the need for and develop the ability to engage in

7	continuous learning as a Computing professional.
PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO 9	Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economic, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO 11	Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.
PO 12	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

Program Specific Outcomes (PSO's)	
PSO1	Explore and analyse the basics of Information Security and Android Development framework.
PSO2	Design and analyse the user interfaces based on the specifications.
PSO3	Recognise the need to use different packages and hardware resources to design a given application.
PSO4	Enterprise and develop high-end mobile applications by considering security aspects.

Course Objectives (COB's)

COB1	To provide the most fundamental knowledge to the students about emerging technologies in computer science
COB2	Explore the current scope, potential, limitations, and implications of EMERGING TECHNOLOGIES
COB3	To demonstrate use cases of emerging technologies in current and futuristic scenarios.

Course Outcomes (CO's)

At the end of the course, students will be able to

Sl. No.	Course Outcome	Description	Bloom's Taxonomy Level
1.	CO1	Fundamental understanding of the cloud computing and its foundation to apply the basic principle of AWS.	L2
2.	CO2	Demonstrate the usage of different machine learning models in real time applications.	L3
3.	CO3	Recognize and explore the basics of AWS Database.	L3
4.	CO4	Exploring AWS Storage and its applications.	L4
5.	CO5	Interpret various concepts of AWS Networking.	L5

CO-PO/PSO Mapping:

Course Outcome	Blooms Taxonomy Levels	Program Outcomes (PO's) Mapped												Program Specific Outcomes (PSO's)			
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1 L2	PS O2 L2	PS O3 L3	PS O4 L4
CO1	L2	3	-	-	-	-	-	3	-	2	-	-	-	3	3	-	-
CO2	L2	3	1	-	-	2	-	3	-	2	-	-	-	3	3	-	-
CO3	L3	3	3	3	2	3	-	3	-	2	-	2	-	2	-	3	2
CO4	L3	3	3	3	3	3	3	3	1	2	-	2	-	-	-	3	3
CO5	L4	3	3	3	3	3	3	3	1	2	-	2	-	-	-	2	2

Course Contents:

Mod	Details	Tools used/	CO	PO
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ule		Assessment and Activity	Mappi ng	Map ping
I	Introduction to Cloud Computing And Amazon Web Services Introduction to Cloud Computing, Cloud Service Delivery Models (IAAS, PAAS, SAAS), Cloud Deployment Models (Private, Public, Hybrid and Community), Cloud Computing Security, Case Study Introduction to Amazon Web Services, Why Amazon? Use Cases, AWS Storage Options, AWS Compute Options, AWS Database Options, AWS Workflow Automation and Orchestration Options, AWS Systems Management And Monitoring Options, AWS Virtual Private Cloud Introduction, Pricing Concepts.	Usage of AWS Portal, Creation of AWS Account.	CO1	PO1,PO2,PO3,P O4,PO5,P O8,P O9,PO10, PO11, PO1
II	Introduction to Ec2: Introduction To EC2, Instance Types And Uses, Auto scaling Instances, Amazon Machine Images (AMIS), Modifying Existing Images, Creating New Images of Running Instances, Converting An Instance Store AMI To An EBS AMI, Instances Backed By Storage Types, Elastic IPS, Elastic Load Balancing.	Exploring AWS EC2 and AMI, tool based learning.	CO2	PO1,PO2,PO3,P O4,PO5,P O8,P O9,PO10, PO11, PO12
III	AWS Database: Introduction to AWS database, What is Amazon RDS? Overview, DB instances, AWS Regions and Availability Zones, Security, Monitoring an Amazon RDS DB instance, How to work with Amazon RDS? How you are charges for Amazon RDS? DB Instances, DB instance storage, Regions, Availability Zones and Local Zones, Multi AZ deployments, DB instance	Exploring AWS Database and AWS RDS, Linked-in Learning Certification.	CO3	PO1,PO2,PO3,P O4,PO5,P O8,P O9,PO10, PO11, PO12

	billing for Amazon RDS, Setting up AWS DB instance.			
IV	AWS Storage and AWS Security: Amazon Storage, S3 Storage Basics, Buckets and Objects, Creating A Web Server Using S3 Endpoints, Managing Voluminous Information with EBS, Glacier Storage Service , Describe Amazon Dynamo DB. Introduction to AWS Security, Describe Amazon Identity and Access Management (IAM), AWS Directory Service, AWS Key Management Service, Securing Data at Rest and In Motion.	Tool based learning-AWS S3, Linked-in Learning Certification.	CO4	PO1,PO2,PO3,P O4,PO5,P O8,P O9,PO10, PO11, PO12
V	AWS Networking: Introduction to AWS Networking , Access Control Lists (ACLs), Setting Up a Security Group, Setting Up VPC And Internet Gateway, Setting Up A VPN, Setting Up A Customer Gateway For VPN, Setting Up Dedicated Hardware For VPC, Scenario 1:VPC With A Public Subnet Only (Standalone Web), Scenario 2: VPC with Public And Private Subnets (3 Tier App), Scenario 3:VPC With Public And Private Subnets And Hardware VPN Access (Web On The Cloud, Database and App On Prem) Scenario 4: VPC With A Private Subnet Only And Hardware VPN Access. (Extension Of Your Corporate Network), Route53 for DNS System, Cloud front, Case Study	Scenario Based Learning,AWS Networking case studies, Technology based puzzles.	CO5	PO1,PO2,PO3,P O4,PO5,P O6,P O7,PO8,P O9,P PO10,PO11,PO12

Text Books (TB):

Textbook and References	
1	Joe Baron, Hisham Baz , Tim Bixler , Biff Gaut , Kevin E. Kelly , Sean Senior , John Stamper , “AWS Certified Solutions Architect Official Study Guide: Associate Exam, John Wiley and Sons Publications, 2017.
2	Yohan Wadia , “AWS Certified Solutions Architect Official Study Guide: Associate Exam, John Packt Publishing, 2016.
3	Bernald Golden, “Amazon Web Services for Dummies”, John Wiley & Sons, 2018
4	Amazon Relational Database Service-User Guide(https://docs.aws.amazon.com)

Other Reading (OR):

OR-1	Introduction to Cloud Computing & AWS	https://aws.amazon.com/what-is-aws/
OR-2	AWS Global Infrastructure	https://aws.amazon.com/about-aws/global-infrastructure/
OR-3	AWS Services Overview	https://aws.amazon.com/products/
OR-4	AWS Well-Architected Framework	https://aws.amazon.com/architecture/well-architected/
OR-1	Introduction to Cloud Computing & AWS	https://aws.amazon.com/what-is-aws/

Relevant Website (RW):

RW No	Topic	Link
RW-1	AWS Beginner Tutorial	https://www.tutorialspoint.com/aws/index.htm
RW-2	AWS Compute Services (EC2, Lambda)	https://aws.amazon.com/ec2/
RW-3	AWS Storage Services (S3, EBS, Glacier)	https://aws.amazon.com/s3/
RW-4	AWS Networking (VPC, Route53)	https://aws.amazon.com/vpc/
RW-5	AWS Security & IAM	https://aws.amazon.com/iam/
RW-6	AWS Database Services (RDS, DynamoDB)	https://aws.amazon.com/products/databases/

Video Links (VL):

VL No	Topic	Link
VL-1	AWS Cloud Computing Full Course	https://www.youtube.com/watch?v=ulprqHHWlng
VL-2	AWS EC2 Tutorial	https://www.youtube.com/watch?v=H8m9HJdS3c0
VL-3	AWS S3 Tutorial for Beginners	https://www.youtube.com/watch?v=4k3Yr_hvN9I
VL-4	AWS IAM Tutorial	https://www.youtube.com/watch?v=2H6W2PKzZ2E

VL-5	AWS VPC & Networking	https://www.youtube.com/watch?v=RytBv1pJ9hM
VL-6	AWS Lambda & Serverless	https://www.youtube.com/watch?v=eOBq_hFvF8
VL-7	AWS RDS Database Tutorial	https://www.youtube.com/watch?v=r6n1-jyS3ak

MOOC Courses (MC):

Sr.No	Platform	Topic	CO	Link	Duration
MC-1	LinkedIn Learning	AWS Essential Training for Beginners	CO1	https://www.linkedin.com/learning/aws-essential-training-for-beginners	3 hr 15 min
MC-2	Coursera	AWS Fundamentals: Going Cloud-Native	CO1, CO2	https://www.coursera.org/learn/aws-fundamentals-going-cloud-native	4 hr
MC-3	LinkedIn Learning	AWS for Developers: Core Services	CO2, CO3	https://www.linkedin.com/learning/aws-for-developers-core-services	2 hr 30 min
MC-4	Udemy	AWS Certified Solutions Architect – Associate	CO3, CO4, CO5	https://www.udemy.com/course/aws-certified-solutions-architect-associate/	15 hr
MC-5	LinkedIn Learning	AWS Security Best Practices	CO4	https://www.linkedin.com/learning/aws-security-best-practices	1 hr 45 min
MC-6	Coursera	Serverless Machine Learning with AWS	CO5	https://www.coursera.org/learn/serverless-ml-aws	5 hr 30 min

Session-wise Planning:

Session	Hours	Topics Covered	Pedagogy / Activity	CO	Mode
1	1	Introduction to Cloud Computing, Evolution & Need	PPT + Video Discussion	CO1	Synchronous
2	1	Cloud Service Models – IaaS, PaaS, SaaS	PPT + Examples	CO1	Synchronous
3	1	Cloud Deployment Models – Public, Private, Hybrid, Community	Case-based explanation	CO1	Synchronous
4	1	Cloud Computing Security – Challenges & Controls	PPT + Discussion	CO2	Synchronous
5	1	Cloud Computing Case Study	Case analysis	CO2	Synchronous
6	1	Introduction to Amazon Web Services	PPT + Demo	CO1	Synchronous
7	1	Why Amazon? AWS Use Cases	Industry use cases	CO2	Synchronous
8	1	AWS Core Services Overview (Storage, Compute, Database)	PPT + AWS Console Demo	CO2	Blended
9	1	AWS VPC Introduction & Pricing Concepts	Demo + Cost calculator	CO2	Blended
10	1	Introduction to Amazon EC2	PPT + Console Overview	CO2	Synchronous
11	1	EC2 Instance Types and Use Cases	PPT + Comparison table	CO2	Synchronous
12	1	Amazon Machine Images (AMIs)	Demo	CO3	Blended
13	1	Creating and Modifying AMIs	Hands-on Lab	CO3	Practical
14	1	Instance Storage vs EBS-backed	PPT + Demo	CO3	Blended

		Instances			
15	1	Converting Instance Store AMI to EBS AMI	Demonstration	CO3	Practical
16	1	Auto Scaling Instances	Demo + Discussion	CO3	Blended
17	1	Elastic IPs	Hands-on	CO3	Practical
18	1	Elastic Load Balancing	Demo + Architecture Diagram	CO3	Practical
19	1	Introduction to AWS Databases	PPT	CO3	Synchronous
20	1	Amazon RDS – Overview & Features	PPT + Demo	CO3	Blended
21	1	DB Instances, Storage Types	PPT + Comparison	CO3	Synchronous
22	1	AWS Regions, AZs, Local Zones	PPT + Architecture	CO3	Synchronous
23	1	Multi-AZ Deployments	Demo	CO3	Practical
24	1	RDS Security	PPT + Demo	CO4	Blended
25	1	Monitoring RDS Instances	CloudWatch Demo	CO4	Practical
26	1	RDS Billing and Pricing	Cost estimation	CO4	Synchronous
27	1	Setting up AWS DB Instance	Hands-on Lab	CO4	Practical
28	1	Amazon Storage Overview	PPT	CO3	Synchronous
29	1	S3 Storage Basics	Demo	CO3	Blended
30	1	Buckets and Objects	Hands-on	CO3	Practical
31	1	Creating Web Server Using S3 Endpoints	Hands-on Lab	CO4	Practical
32	1	EBS – Managing Voluminous Data	PPT + Demo	CO4	Blended
33	1	Glacier Storage Service	PPT + Use case	CO4	Synchronous
34	1	Amazon DynamoDB	PPT + Demo	CO4	Blended
35	1	Introduction to AWS Security	PPT	CO4	Synchronous
36	1	IAM, KMS, Securing Data at Rest & In Motion	Demo	CO4	Practical
37	1	Introduction to AWS Networking	PPT	CO3	Synchronous
38	1	ACLs & Security Groups	Demo	CO4	Blended
39	1	Setting up VPC & Internet Gateway	Hands-on Lab	CO4	Practical
40	1	VPN & Customer Gateway	Demo	CO4	Practical
41	1	Dedicated Hardware for VPC	PPT + Architecture	CO4	Synchronous
42	1	Scenario 1: VPC with Public Subnet	Case-based Lab	CO5	Practical
43	1	Scenario 2: Public & Private Subnets (3-Tier)	Case-based Lab	CO5	Practical
44	1	Scenario 3 & 4: Hybrid VPC Architectures	Architecture Design	CO5	Practical
45	1	Route53, CloudFront & Case Study	PPT + Discussion	CO5	Blended

Assessment Scheme

Internal Assessment (IA) : End Semester Examination (ESE) = 50 : 50

Table 1: Assessment Instruments and Weightage

Sl.	Assessment Instrument	Formative /	Frequency	Weightage	CO Mapping
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No.		Summative		(%)	
1	Activity–1	Formative	1	15	CO1, CO2, CO3, CO4, CO5
2	Activity–2	Formative	1	15	CO1, CO2, CO3, CO4, CO5
3	Internal Test–1	Summative	1	5	CO1, CO2
4	Internal Test–2	Summative	1	10	CO1, CO2, CO3, CO4, CO5
5	Class Participation	Formative	Continuous	5	CO1, CO2, CO3, CO4, CO5
6	End Semester Examination (ESE)	Summative	1	50	CO1, CO2, CO3, CO4, CO5

Notes and Guidelines

1) MOOC (NPTEL / LinkedIn Learning / EdX)

- Students must submit the **course completion certificate** along with a **written report**, duly signed by the faculty.
- The report will be evaluated, and a **Viva Voce** examination shall be conducted.

2) Choice-Based Faculty Activity

Faculty may choose **any one** of the following based on subject requirements:

- Seminar
- Group Discussion
- Debate
- Poster Presentation
- Interview
- Project
- Case Study
- Book Chapter Review / Interpretation
- Research Paper Interpretation
- Report Writing
- Field Study
- Data Analysis
- Model / Architecture Development
- Database Design
- User Interface / Sequence Design

- Students must submit a **written report**, duly signed by the faculty.
- The report will be evaluated, and a **Viva Voce** shall be conducted.

3) Periodic Internal Tests

- One internal examination** is conducted in each semester, **after 4 weeks** of instruction.
- Syllabus coverage per internal test: **2 to 2.5 units (30%–40%)**.
- Tests are **centralized and conducted through LMS**.
- Each question is mapped with the **Course Outcome (CO)** number.
- Evaluated answer scripts are returned to students, and **valuation grievances**, if any, are addressed by the teacher.

4) Preparatory Examination

- One preparatory examination is conducted in each semester.
- Syllabus coverage includes **all five modules**.

Assessment Sheet with Rubrics for Grading & Evaluation

Activity–1 (A1): Online Certification Course (12–15 Hours)

Sr. No.	USN No.	Student Name	Presentation (10)	On-time Submission (10)	Conclusion with Learning Outcomes (20)	Originality of Report ($\leq 12\%$) (10)	Total (50)	Converted Marks (15)

Activity–2 (A2): PowerPoint Presentation on Mobile Devices Application Testing

Sr. No.	USN No.	Student Name	On-time Submission (10)	Abstract Submission (10)	Testing Case Study Review (10)	Report with Conclusion (10)	Originality of Report ($\leq 12\%$) (10)	Total (50)	Converted Marks (15)